SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 6.01

C SAAT Rational Functions, Sequences, Series & Logarithms

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

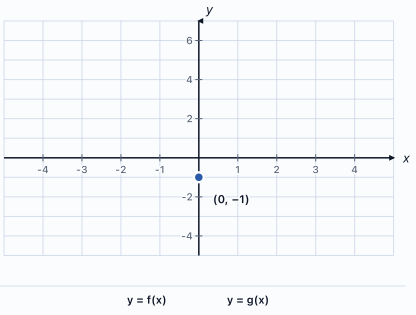
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 B	Q2 C	Q3 D	Q4 A	Q5 B
Q6 C	Q7 D	Q8 A	Q9 D	Q10 A
Q11 C	Q12 B	Q13 C	Q14 B	Q15 A
Q16 B	Q17 D	Q18 B	Q19 A	Q20 D
Q21 B	Q22 C	Q23 D	Q24 D	Q25 D

SAAT Rational Functions, Sequences, Series & Logarithms Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 16x + 80$ in the domain of SAAT Functions, Sequences & Logarithms. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(10)$.



A 44

B 36 ✓

C 20

D 34

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Functions, Sequences & Logarithms):

1. Complete the square: $x^2 - 16x + 80 = (x-8)^2 - 64 + 80 = (x-8)^2 + 16$, giving minimum value $k = 16$. **[M1]**
2. Evaluate at $x = 10$: $f_1(10) = (2)^2 + 16 = 20$. **[M1]**
3. Sum the two quantities: $k + f_1(10) = 16 + 20 = 36$. **[A1]**

- **Final Verified Answer:** 36.

2 MCQ

A mathematical model in SAAT Functions, Sequences & Logarithms is defined by the cubic function $y_2(x) = 4x^3 - 8x^2 + 15x - 13$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 123

B 60

C 75 ✓

D 35

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Functions, Sequences & Logarithms):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 12x^2 - 16x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 12(9) - 16(3) + 15 = 75$. **[M1][A1]**

- **Final Verified Answer:** 75.

3 MCQ

In SAAT Functions, Sequences & Logarithms, two displacement vectors are given by $\mathbf{u}_3 = \langle 8, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A 16

B 128

C 256

D 144 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Functions, Sequences & Logarithms):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (8)(9) + (-8)(7) = 72 + (-56) = 16$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 8^2 + (-8)^2 = 128$. **[M1]**
3. Combine: $16 + 128 = 144$. **[A1]**

- **Final Verified Answer:** 144.

4 MCQ

The linear growth profile shown in the diagram for SAAT Functions, Sequences & Logarithms follows an arithmetic progression with first term $a_1 = 9$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 765 ✓

B 93

C 855

D 1395

WORKED SOLUTION & VERIFICATION

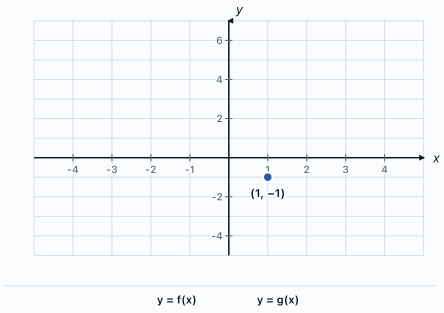
Mark Scheme & Examiner Solution (SAAT Functions, Sequences & Logarithms):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 9, d = 6$: $S_{15} = \frac{15}{2}[18 + 14 \times 6] = \frac{15}{2} \times 102 = 765$. **[M1][A1]**

- **Final Verified Answer:** 765.

5 MCQ

In SAAT Functions, Sequences & Logarithms, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 32x)\,\mathrm{d}x$.



A 136

B 112 ✓

C 144

D 57

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Functions, Sequences & Logarithms):

1. Integrate term by term: $\int (18x^2 + 32x)\,\mathrm{d}x = [6x^3 + 16x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 16(2)^2) - 0 = 48 + 64 = 112$. **[M1][A1]**

- **Final Verified Answer:** 112.

6 MCQ

In SAAT Functions, Sequences & Logarithms, consider the quadratic function $f_6(x) = x^2 - 18x + 169$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(16)$.

A 234

B 137

C 225 ✓

D 218

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Functions, Sequences & Logarithms):

1. Complete the square: $x^2 - 18x + 169 = (x-9)^2 - 81 + 169 = (x-9)^2 + 88$, giving minimum value $k = 88$. **[M1]**
2. Evaluate at $x = 16$: $f_6(16) = (7)^2 + 88 = 137$. **[M1]**
3. Sum the two quantities: $k + f_6(16) = 88 + 137 = 225$. **[A1]**

- **Final Verified Answer:** 225.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 4x^3 - 9x^2 + 33x - 20$ studied in SAAT Functions, Sequences & Logarithms. Find the instantaneous gradient $\frac{\mathrm{dy}}{\mathrm{dx}}$ of the tangent to the curve at $x = 8$.

A 801

B 624

C 111

D 657 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Functions, Sequences & Logarithms):

1. Differentiate term by term using the power rule: $\frac{\mathrm{dy}}{\mathrm{dx}} = 12x^2 - 18x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{\mathrm{dy}}{\mathrm{dx}}\right|_{x=8} = 12(64) - 18(8) + 33 = 657$. **[M1][A1]**

- **Final Verified Answer:** 657.

8 MCQ

In SAAT Functions, Sequences & Logarithms, two displacement vectors are given by $\mathbf{u}_8 = \langle 9, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

- **A** 167 ✓
- **B** 37
- **C** 130
- **D** 237

WORKED SOLUTION & VERIFICATION

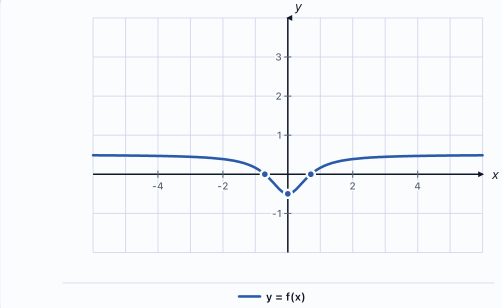
Mark Scheme & Examiner Solution (SAAT Functions, Sequences & Logarithms):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (9)(8) + (-7)(5) = 72 + (-35) = 37$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 9^2 + (-7)^2 = 130$. **[M1]**
3. Combine: $37 + 130 = 167$. **[A1]**

- **Final Verified Answer:** 167.

9 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{2x^2 - 3x + 1}{4x^2 + 2}$.



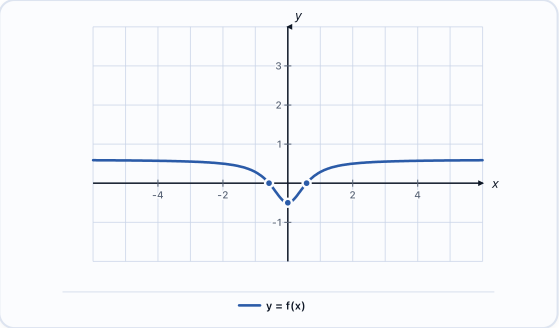
- **A** $y = 0$
- **B** $y = 2$
- **C** $y = 6$
- **D** $y = \frac{1}{2}$ ✓

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 2, the horizontal asymptote is the ratio of leading coefficients $y = \frac{2}{4} = \frac{1}{2}$.

10 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{3x^3-4x+1}{5x^3+3}$.



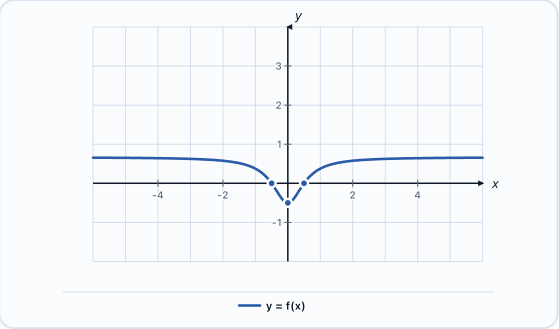
- **A** $y = \frac{3}{5}$ ✓
- **B** $y = 0$
- **C** $y = \frac{5}{3}$
- **D** $y = 8$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 3, the horizontal asymptote is the ratio of leading coefficients $y = \frac{3}{5} = \frac{3}{5}$.

11 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{4x^4-5x+1}{6x^4+4}$.



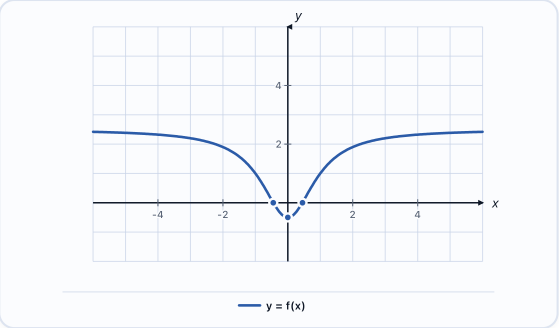
- **A** $y = 10$
- **B** $y = 0$
- **C** $y = \frac{2}{3}$ ✓
- **D** $y = \frac{3}{2}$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 4, the horizontal asymptote is the ratio of leading coefficients $y = \frac{4}{6} = \frac{2}{3}$.

12 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{5x^2-6x+1}{2x^2+5}$.



A $y = 0$

B $y = \frac{5}{2}$ ✓

C $y = \frac{2}{5}$

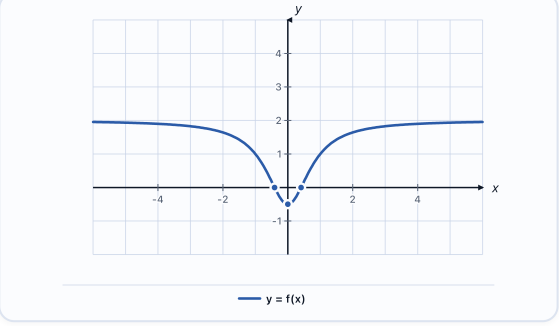
D $y = 7$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 2, the horizontal asymptote is the ratio of leading coefficients $y = \frac{5}{2} = \frac{5}{2}$.

13 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{6x^3-7x+1}{3x^3+6}$.



A $y = 0$

B $y = \frac{1}{2}$

C $y = 2$ ✓

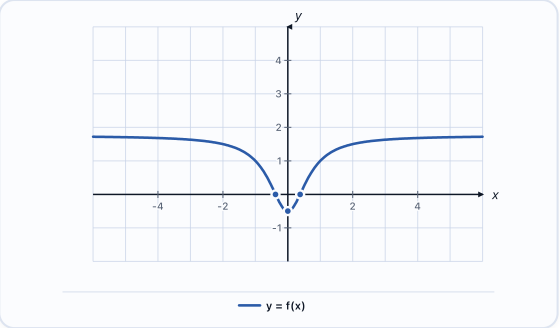
D $y = 9$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 3, the horizontal asymptote is the ratio of leading coefficients $y = \frac{6}{3} = 2$.

14 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{7x^4-8x+1}{4x^4+7}$.



A $y = 0$

B $y = \frac{7}{4}$ ✓

C $y = \frac{4}{7}$

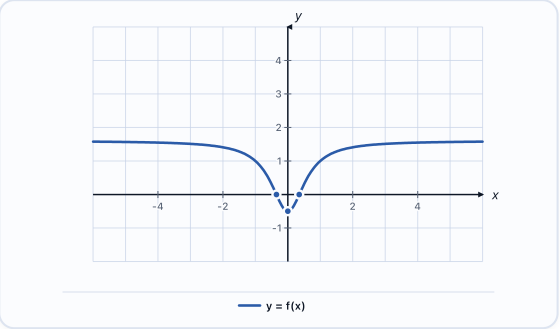
D $y = 11$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 4, the horizontal asymptote is the ratio of leading coefficients $y = \frac{7}{4} = \frac{7}{4}$.

15 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{8x^2-9x+1}{5x^2+8}$.



A $y = \frac{8}{5}$ ✓

B $y = 13$

C $y = 0$

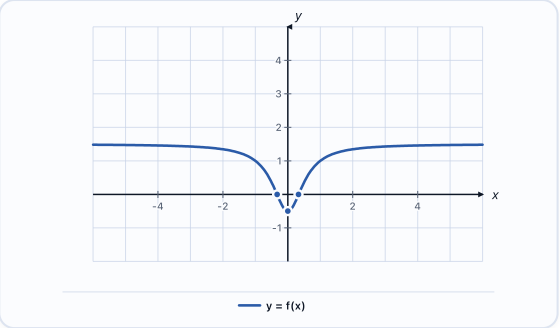
D $y = \frac{5}{8}$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 2, the horizontal asymptote is the ratio of leading coefficients $y = \frac{8}{5} = \frac{8}{5}$.

16 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{9x^3-10x+1}{6x^3+9}$.



A $y = 15$

B $y = \frac{3}{2}$ ✓

C $y = 0$

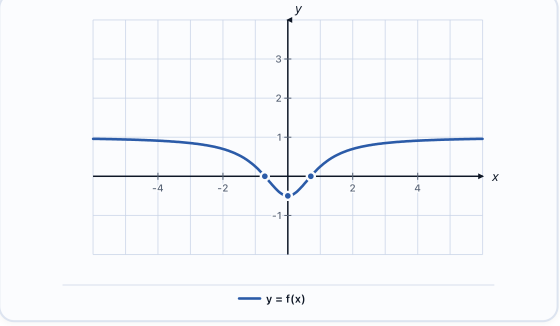
D $y = \frac{2}{3}$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 3, the horizontal asymptote is the ratio of leading coefficients $y = \frac{9}{6} = \frac{3}{2}$.

17 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{2x^4-11x+1}{2x^4+10}$.



A 14

B $y = 0$

C $y = 4$

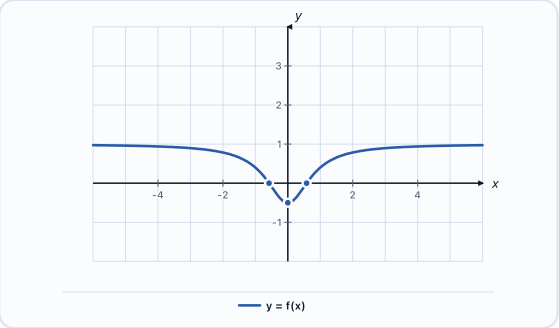
D $y = 1$ ✓

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 4, the horizontal asymptote is the ratio of leading coefficients $y = \frac{2}{2} = 1$.

18 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{3x^2-12x+1}{3x^2+11}$.



A 14

B $y = 1$ ✓

C $y = 0$

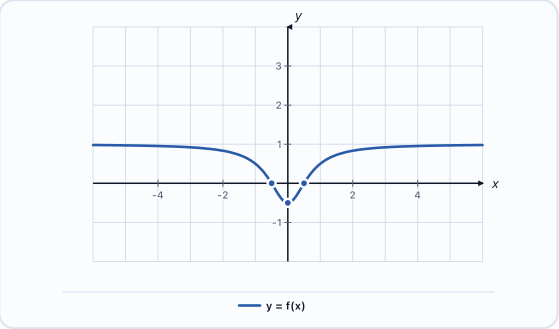
D $y = 6$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 2, the horizontal asymptote is the ratio of leading coefficients $y = \frac{3}{3} = 1$.

19 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{4x^3-13x+1}{4x^3+12}$.



A $y = 1$ ✓

B 14

C $y = 0$

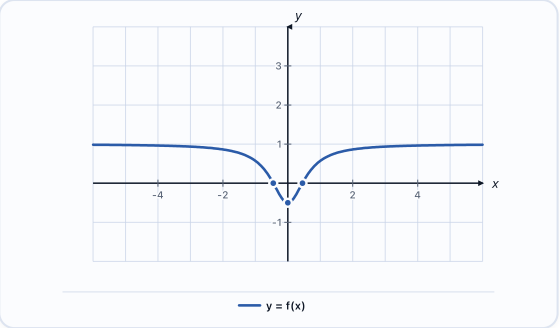
D $y = 8$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 3, the horizontal asymptote is the ratio of leading coefficients $y = \frac{4}{4} = 1$.

20 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{5x^4-14x+1}{5x^4+13}$.



A $y = 10$

B 14

C $y = 0$

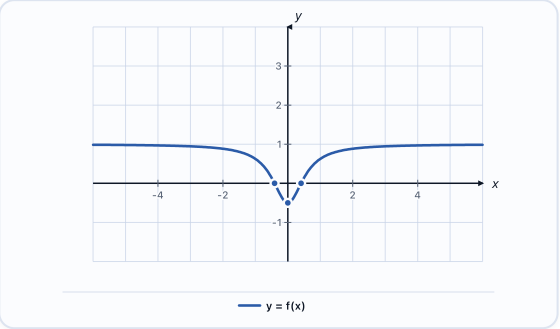
D $y = 1$ ✓

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 4, the horizontal asymptote is the ratio of leading coefficients $y = \frac{5}{5} = 1$.

21 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{6x^2-15x+1}{6x^2+14}$.



A $y = 0$

B $y = 1$ ✓

C $y = 12$

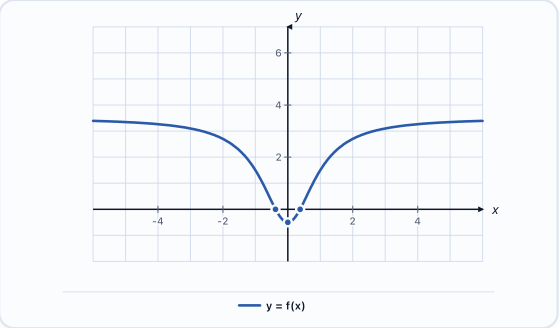
D 14

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 2, the horizontal asymptote is the ratio of leading coefficients $y = \frac{6}{6} = 1$.

22 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{7x^3-16x+1}{2x^3+15}$.



A $y = 0$

B $y = \frac{2}{7}$

C $y = \frac{7}{2}$ ✓

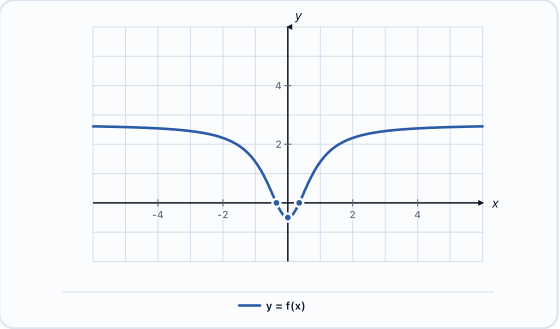
D $y = 9$

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 3, the horizontal asymptote is the ratio of leading coefficients $y = \frac{7}{2} = \frac{7}{2}$.

23 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{8x^4-17x+1}{3x^4+16}$.



A $y = 11$

B $y = \frac{3}{8}$

C $y = 0$

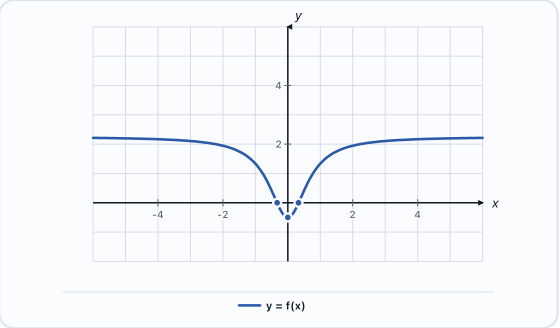
D $y = \frac{8}{3}$ ✓

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 4, the horizontal asymptote is the ratio of leading coefficients $y = \frac{8}{3} = \frac{8}{3}$.

24 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{9x^2-18x+1}{4x^2+17}$.



A $y = 13$

B $y = 0$

C $y = \frac{4}{9}$

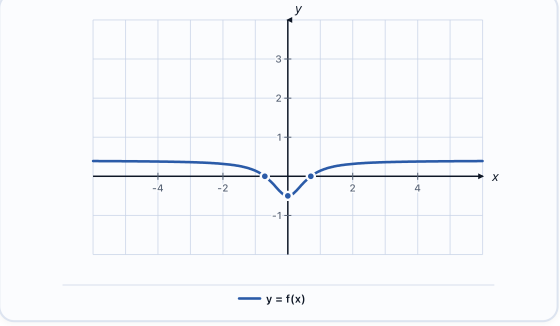
D $y = \frac{9}{4}$ ✓

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 2, the horizontal asymptote is the ratio of leading coefficients $y = \frac{9}{4} = \frac{9}{4}$.

25 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{2x^3-19x+1}{5x^3+18}$.



A $y = 7$

B $y = \frac{5}{2}$

C $y = 0$

D $y = \frac{2}{5}$ ✓

WORKED SOLUTION & VERIFICATION

Because numerator and denominator have equal degree 3, the horizontal asymptote is the ratio of leading coefficients $y = \frac{2}{5} = \frac{2}{5}$.